INCH-POUND

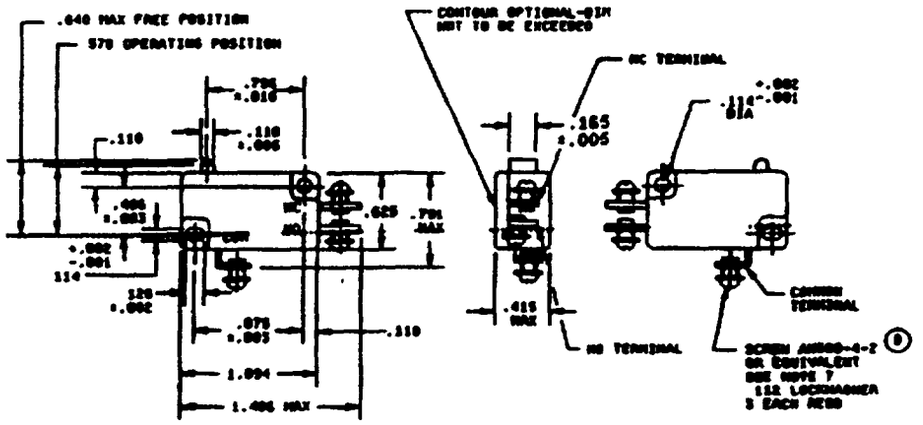
MIL-S-8805/90D(EC)
6 June 1995
SUPERSEDING
MIL-S-8805/90C(EC)
6 January 1971

MILITARY SPECIFICATION SHEET

(D) SWITCH, SENSITIVE, LOW FORCE, SPDT, 10 AMPERES, UNSEALED

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.




| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .001 | 0.03 | .110 | 2.79 | .415 | 10.54 | .796 | 20.22 |
| .002 | 0.05 | .114 | 2.90 | .578 | 14.68 | .875 | 22.23 |
| .003 | 0.08 | .126 | 3.20 | .625 | 15.88 | 1.094 | 27.79 |
| .005 | 0.13 | .165 | 4.19 | .640 | 16.26 | 1.406 | 35.71 |
| .010 | 0.25 | .406 | 10.31 | .791 | 20.09 | | |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.015 inch (0.38 mm).
3. Exact shape of switch optional provided dimensions specified are not exceeded.
4. Mark, stamp, or mold NO, NC, and COM. terminal designation on either side of switch
5. Metric equivalents are given for general information only.
6. Mounting holes shall accept .112 (2.82 mm) maximum diameter pins or screws.
7. (D) Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for terminal screws and lock washer material. Dimensions shall be in accordance with the referenced hardware specifications.

FIGURE 1. Configuration and dimensions.

(D) denotes changes



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Shock: 50 g's, MIL-STD-202, method 213, test condition A.

Vibration grade: 1 (10 to 500 Hz).

Weight: 0.30 ounce maximum.

Terminal strength: 15 pounds.

Operating characteristics:

  Actuating force: 2.65 ounces maximum.

  Pre travel: .050 inch (1.27 mm) maximum.

  Movement differential: .010 inch (0.25 mm) maximum.

  Overtravel: .050 inch (1.27 mm) minimum.

Strength of actuating means: 25 pounds.

Material (Case): Diallyl phthalate (in accordance with MIL-M-14).

Contact resistance: Not applicable.

Dielectric withstanding voltage:

  Sea level: 1,000 V rms.

  Altitude: 400 V rms at 50,000 feet.

Mechanical endurance: 25,000 cycles.

Electrical endurance: 25,000 cycles.

  Electrical ratings: See table I.

Qualification:

  Group submission: See table II.

Part or Identifying Number (PIN): See table III.



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TABLE I. Electrical ratings.

| Load | Sea level | | 50,000 feet | |
|---|---|---|---|---|
| | 28 V dc | 115 V ac, 60 Hz | 28 V dc | 115 V ac, 60 Hz |
| | Amperes | Amperes | Amperes | Amperes |
| Resistive | 10 | 10 | 10 | 10 |
| Inductive | 6 | 10 | 6 | 10 |
| Motor | 6 | 3 | --- | --- |

TABLE II. Group submission.

| Examination or test | Basic switch | Other switch samples | Extent of approval |
|---|---|---|---|
| Qualification inspection table of MIL-S-8805, groups I through VIII, inclusive | M8805/90-001 (all sample units) | M8805/90-002 and -003 (2 sample units each) visual and mechanical examination | All |

TABLE III. PIN's and characteristics.

| PIN's M8805/90 | Terminals | Switching characteristics |
|---|---|---|
| -001 | NC, NO and Common | SPDT |
| -002 | NO and Common | SPNO |
| -003 | NC and Common | SPNC |

CONCLUDING MATERIAL

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-N742)



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